NATIONAL INSTITUTE OF TECHNOLOGY KURUKSHETRA

DEPARTMENT OF CHEMISTRY

Examination: **B. Tech. 1st Year 2nd Semester (Midterm-I Exam.)**

Date of Examination: **21.02.2025** **Timing: 8:00 am to 8.50 am**

Subject & Code: **Chemistry (CHIC- 102) For CE, ME & PIE Branches**

Maximum Marks: 15

Note: i) Attempt all Questions. ii) All Questions carry equal marks.
iii) Additional Answer book will not be provided.

Q1. Draw the structure / representation of :
ii) A graft polymer ii) Isotactic Propylene iii) A thermosetting polymer (1x3)

Q2. Polymerisation of ethylene gives different structural polyethylenes under different reaction conditions. Explain. **OR** (3)

Give the chemical reactions for preparing Polydimethyl silicone (PDMS) and its conversion to vulcanized silicone rubber. (2,1)

Q3. Give two disadvantages of scale formation in boilers. How scale formation can be prevented by calgon conditioning? (1,2)

Q4. Draw the chemical structure of a cation exchange resin. Give the ion exchange reactions involved in water softening using it and also for its regeneration. (1,1,1)

Q5. a) Describe the advantages of reverse osmosis process for desalination of water.
b) Give two technical applications of Buna-N. (2,1)